



Deep Learning 1: exam

Duration : 105 minutes (1h45). (do not spend too much time on the MCQ)

Important: Strictly forbidden: books, laptops, phones, ...

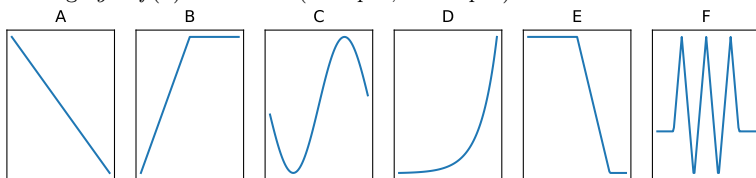
Important: Each MCQ question have at least one correct answer and may have multiple correct answers (check all that are correct). The answer "(other)" must be checked only when none of the other answers is correct.

Question 1 About you:

LASTNAME (NOM)

FIRSTNAME (PRÉNOM)

Given the following " $y = f(x)$ " functions (1d input, 1d output).



Question 2 Which(s) function(s) can be arbitrarily well approximated on the shown interval by a perceptron with a ReLU activation function.

☐ C ☐ D ☒ A ☒ B ☐ (other) ☐ F ☐ E

Question 3 Which(s) function(s) can be arbitrarily well approximated on the shown interval by a wide MLP with one hidden layer of as many neurons as we want, with a ReLU activation (on this layer only).

☒ C ☒ A ☒ D ☒ F ☐ (other) ☒ E ☒ B

Question 4 Which(s) function(s) can be arbitrarily well approximated on the shown interval by a perceptron with a sigmoid activation function.

☒ A ☐ E ☐ D ☐ F ☐ B ☐ (other) ☐ C

Question 5 Which(s) function(s) can be arbitrarily well approximated on the shown interval by a wide MLP with three hidden layers of as many neurons as we want, with no activations.

☐ D ☐ (other) ☐ B ☐ C ☐ E ☐ F ☒ A

$w_3(w_2(w_1x+b_1)+b_2)+b_3$

doesn't matter the 3 layers

Question 6 Given a numpy array X with shape $(10,9,21,33,42)$, and

$W = X[0,2:4:6,::8,-10,:]$,

`print(np.shape(W))` then outputs $1,3,42$

Question 7 If X has shape $(4,5,1)$ and W has shape $(1,5,9)$, we have to match each dimension

`print(np.shape(X * W))` prints $[4,5,9]$

(write ERROR in case of error) as long as one dimension has size 1 when mismatch, ok

Question 8 We have loaded a dataset file as a numpy array J with 100 points (lines) and 10 columns, the first column is the label. How can we use numpy to extract and save into a variable X the "training points", that is the first 80 lines and all columns except the first one.

$X = J[80, 1:]$

Question 9 An MLP with one layer of 1000 hidden neurons, taking as input a vector of length 10, producing an output of length 4 has approximately that many parameters

☐ 100000 ☒ 15000 ☐ 10000 ☐ 1004 ☐ 1014

Question 10 An MLP taking as input a vector of length 2, having 15 layers each having 2 hidden neurons and producing an output of length 2 has approximately that many parameters

☒ 100 ☐ 2^{16} ☐ 15 ☐ 6^{16}

Question 11 A simple perceptron doing image segmentation (binary), working on 1000×1000 images (and producing a 1000×1000 probability map) has approximately that many parameters

☐ much less ☒ 10^{12} ☐ 10^9 ☐ 10^3 ☐ 1

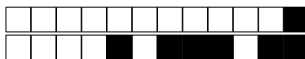
a perceptron for each pixel
so 1000×1000 inputs and outputs
 $(1000 \times 1000)^2$ for i/o

Question 12 A convolutional network with a single convolutional layer doing image segmentation (binary), working on 1000×1000 images (and producing a 1000×1000 probability map) has approximately that many parameters

☐ 10^6 ☐ much more ☐ 10^9 ☒ 10^3 ☐ 10^{12} ☐ much less

Question 13 If we have $H = h(G)$ with $G = g(F)$ with $F = f(X)$, using a combination of some of the following elements: $A: \frac{\partial G}{\partial H}$; $B: \frac{\partial G}{\partial F}$; $C: \frac{\partial G}{\partial X}$; $D: \frac{\partial X}{\partial G}$; $E: \frac{\partial H}{\partial G}$; $M: \frac{\partial F}{\partial X}$; $N: \frac{\partial F}{\partial G}$; $F: (F)$; $G: (G)$; $H: (H)$; $X: (X)$; $\cdot$: multiplication; $/$: division; what is the expression for?

$\frac{\partial H}{\partial X}(X) = E \cdot B \cdot M$ (use the provided letters)



+1/2/59+

Question 14 If we have $H = h(G)$ with $G = g(F)$ with $F = f(X)$, using a combination of some of the following elements: **A**: $\frac{\partial G}{\partial H}$; **B**: $\frac{\partial G}{\partial F}$; **C**: $\frac{\partial G}{\partial X}$; **D**: $\frac{\partial X}{\partial G}$; **E**: $\frac{\partial H}{\partial G}$; **M**: $\frac{\partial F}{\partial X}$; **N**: $\frac{\partial F}{\partial G}$; **F**: (F) ; **G**: (G) ; **H**: (H) ; **X**: (X) ; **.**: multiplication; **/**: division; what is the expression for?

$\frac{\partial G}{\partial X}(X) =$ (use the provided letters)

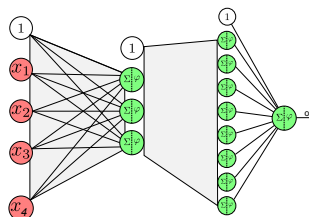
Question 15 In the *pancake* framework (similarly in PyTorch), defining a new layer or module involves writing a `compute_forward` method/function. This python function computes the output of the mathematical function, given its inputs and the parameters stored in the class. Explain with your own words and examples, what the other method/function, `compute_backward`, is doing? What does this function receive as input? What does it compute? Does it call or interact with `compute_forward`?

Question 16 ... (following the previous question)
Explain with your own words (and give pseudo code when possible) how these are used to find good parameters?

Question 17 Explain what you did and summarize the key points you remember/learned from practical session 1.

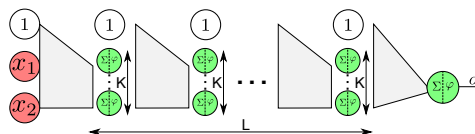
Question 18 Explain one or two of the most complicated/surprising/original experiments you tested in practical session 3 (or outside of the practical session, but then precise the context). What conclusions did you draw from these experiments?

Question 19 Give the expression (and compute it or approximate if possible) for the number of parameters of this MLP.

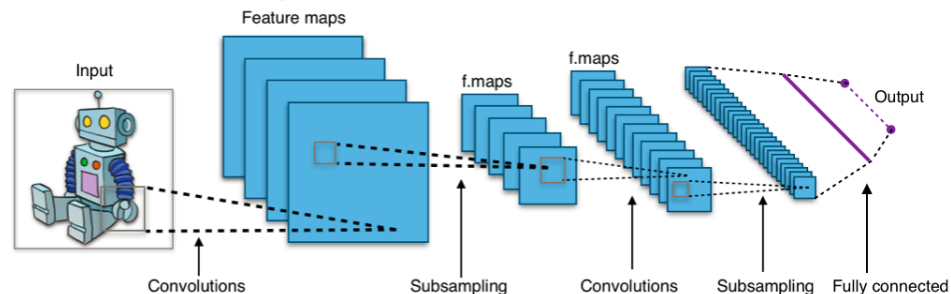


Question 20

Give the expression (and compute it or approximate if possible) for the number of parameters of this MLP.



Question 21 Given this picture of a "convnet".



Explain the architecture by:

- Explaining what each layer does.
- Specifying (proposing meaningful values for) the elements that are unspecified in the picture.
- Giving the sizes of the intermediate tensors that are computed.
- Giving a detail and the (expression of the) count of the parameters.